AIML Engineering Lab

# The Linear Regression Engine

*From first-principles OLS to production-grade regularized models — applied to food science and semiconductor engineering.*

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# PROJECT 1

Artisan Cheese Fermentation — Predicting Optimal Fermentation Time

## 1 Overview

**Objective:** Build a complete linear regression pipeline from first principles — covering Ordinary Least Squares (OLS), Ridge (L2), Lasso (L1), and Elastic Net regularization — to predict the optimal fermentation time of artisan cheese.

**Domain:** Food Science & Fermentation Engineering

**Target Variable:** Optimal fermentation time (hours)

This project implements linear regression from scratch using the Normal Equation and Gradient Descent, then compares against scikit-learn's optimized implementations. The pipeline progresses from basic OLS through regularized variants, demonstrating why regularization prevents overfitting and how L1 vs L2 penalties differ geometrically.

### Key Concepts

- **OLS (Normal Equation)** — Closed-form solution using the pseudo-inverse, efficient for small-to-medium datasets.
- **Gradient Descent** — Iterative optimization that minimizes MSE by following the negative gradient.
- **Ridge (L2)** — Adds squared penalty to coefficients, shrinking all toward zero but never eliminating features.
- **Lasso (L1)** — Adds absolute penalty, driving some coefficients to exactly zero for automatic feature selection.
- **Elastic Net** — Combines L1 and L2 penalties for a balanced regularization approach.

## 2 Datasets

### Artisan Cheese Fermentation Data

| Property | Value |
|---|---|
| **Records** | 2,000 samples |
| **Features** | 8 (7 numeric + 1 categorical) |
| **Target** | optimal_fermentation_time (8 – 73 hours) |
| **Split** | Train 1,400 · Validation 300 · Test 300 |
| **Missing Values** | None |

## Feature Descriptions

| Feature | Description |
|---|---|
| **milk_fat_percentage** | Fat content of the milk |
| **starter_culture_ph** | pH level of the starter culture |
| **ambient_temperature** | Room temperature during fermentation |
| **fermentation_humidity** | Relative humidity in the fermentation room |
| **salt_concentration** | Salt content added to the cheese |
| **curd_cut_size** | Size of the curd cuts |
| **aging_room_airflow** | Air circulation in the aging room |
| **bacterial_strain_type** | Type of bacteria used (categorical — 3 strains) |

## 3 Flowchart

Artisan Cheese Fermentation Pipeline
Data Ingestion
Load artisan_cheese_fermentation_data.csv (2,000 rows); define target = optimal_fermentation_time
Exploratory Data Analysis
Inspect target distribution, feature correlations, and missing values
Data Preprocessing
Standardize numeric features + one-hot encode bacterial_strain_type; split 70/15/15
OLS (From Scratch)
Normal Equation fit (bias + weights) on preprocessed training data
Regularization: Ridge, Lasso, Elastic Net
Train on training set; validate on validation set for shrinkage/selection
Cross-Validation
5-fold CV (RidgeCV, LassoCV) to select alpha minimizing validation error
Test Evaluation
Report RMSE, MAE, R² on held-out test set; overfitting check
Unseen Data Simulation
Generate 500 new samples (seed=999); transform –> predict; compute RMSE/MAE/R²

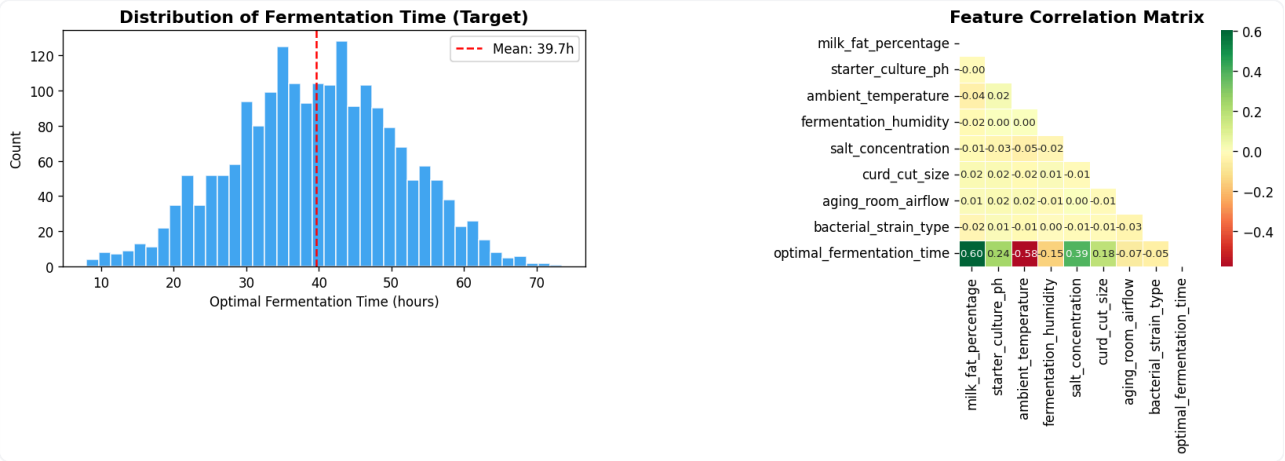
# 4 Results

All models were evaluated on a held-out test set of 300 samples. Ridge (with cross-validated α = 1.15) achieved the best balance of accuracy and generalization.

| Model | RMSE | MAE | $R^2$ |
|---|---|---|---|
| **OLS (Normal Equation)** | 2.441 | 1.931 | 0.9535 |
| **Ridge (α = 1.15)** | 2.442 | 1.931 | 0.9535 |
| **Lasso (α = 0.001)** | 2.442 | 1.931 | 0.9535 |
| **Elastic Net** | 2.604 | 2.057 | 0.9471 |

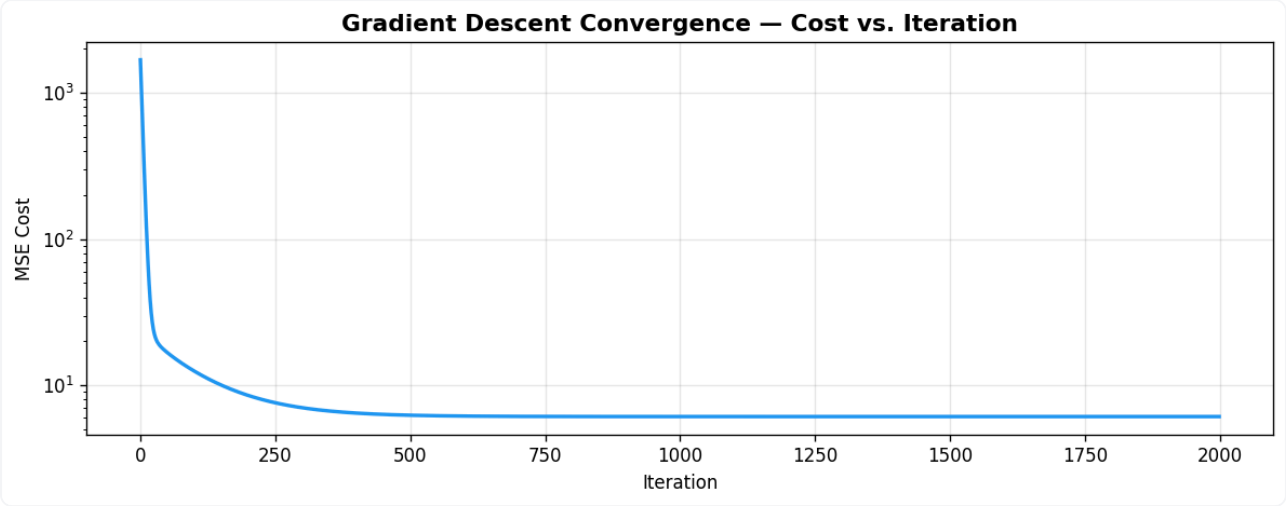
**Best Model:** Ridge (L2) — Test $R^2$ = 0.9535 · RMSE = 2.44 hours — the model explains 95.4% of fermentation time variance.

## 4.1 Exploratory Data Analysis



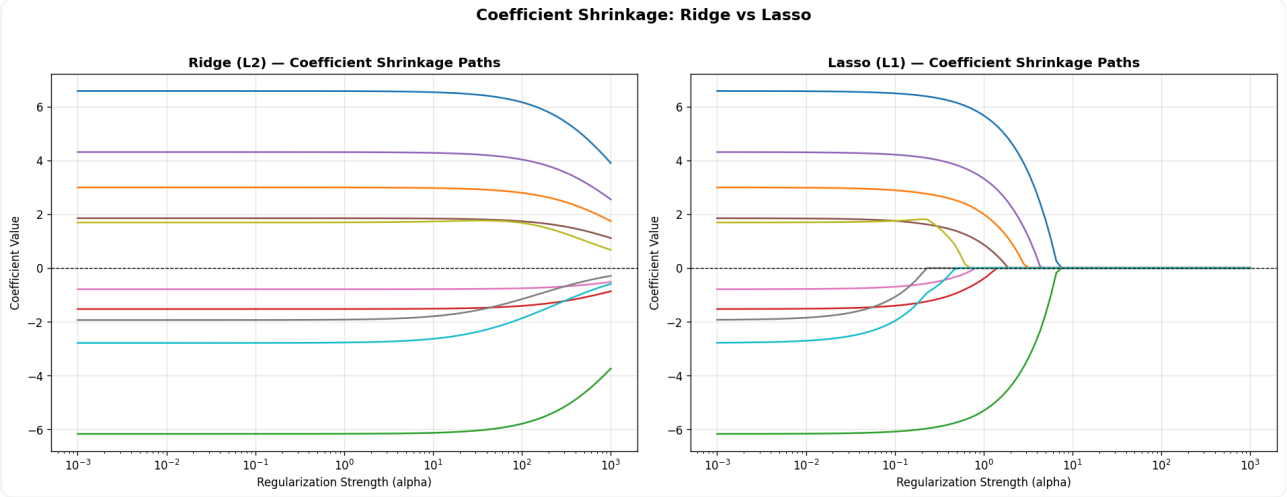

The left panel shows the distribution of optimal fermentation times — a roughly bell-shaped spread between 8 and 73 hours with a central peak around 40 hours, indicating a well-balanced dataset without extreme skew. The right panel is a feature correlation heatmap revealing which biochemical factors carry the strongest linear signal for predicting fermentation time. Strong correlations appear between related measurements (e.g., temperature and humidity), helping guide feature engineering decisions and flagging potential multicollinearity.

## 4.2 Gradient Descent Convergence




The MSE cost function plotted against training iterations on a logarithmic scale. A steep initial drop in the first 100 iterations shows the model quickly capturing the dominant patterns, while the curve flattens around iteration 500 confirming convergence to the global minimum. This validates a well-tuned learning rate — too high would cause oscillation or divergence, too low would require thousands more iterations with no practical improvement.

## 4.3 Coefficient Shrinkage Paths

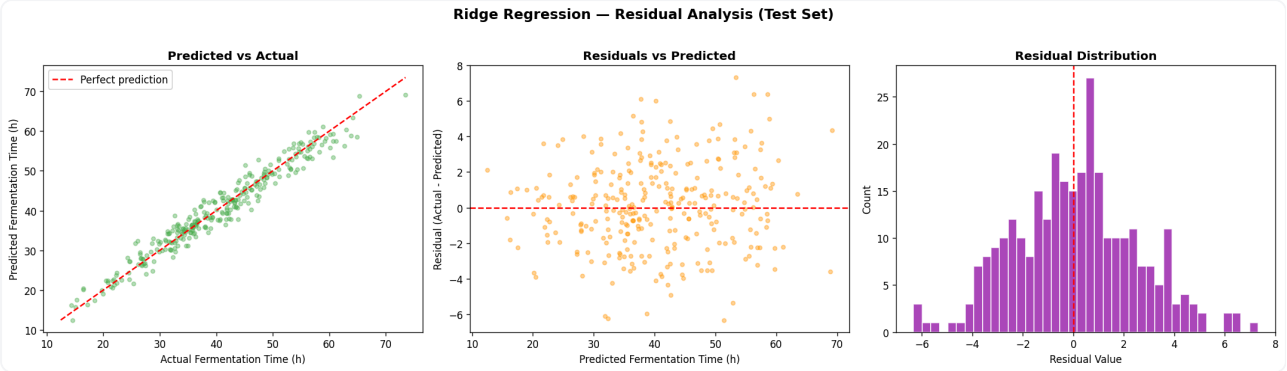



Ridge (left) vs Lasso (right) coefficient paths as penalty strength (alpha) increases from left to right. Under Ridge regularization, all coefficients shrink gradually toward zero but never reach it — every feature contributes to the final prediction. Under Lasso, individual coefficients are driven to exactly zero at different alpha thresholds, effectively removing features from the model. This is the geometric explanation: the L1 diamond constraint has sharp corners on the axes where the optimal solution tends to land, naturally producing sparse models.

## 4.4 Train vs Test Fit

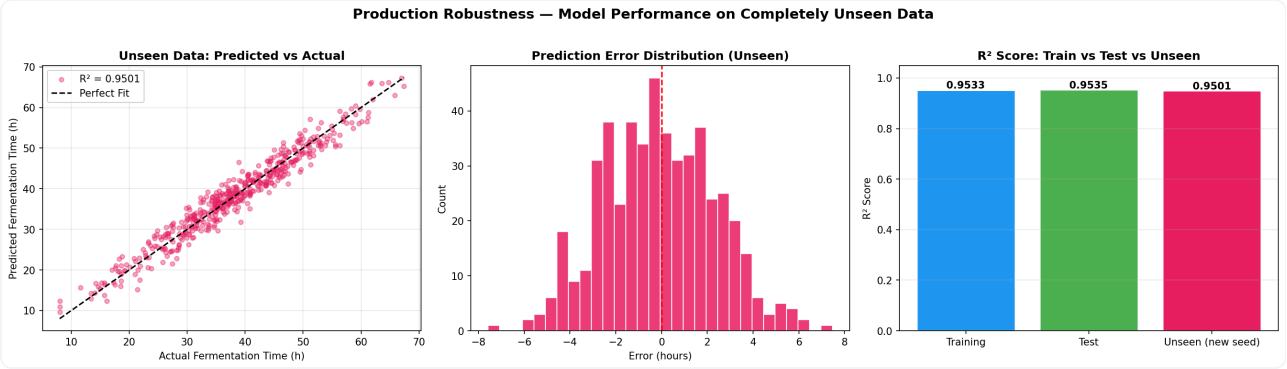
Actual vs predicted fermentation time on both the training set and the held-out test set, with the ideal y = x diagonal shown as reference. Points tightly hugging the line in both panels confirm that the model generalizes well — the consistency between training and test performance indicates no overfitting, and the spread of residuals appears uniform across the entire range of fermentation times.

## 4.5 Residual Analysis




Three-panel residual diagnostic: (1) Predicted vs Actual with the ideal fit line — tight clustering confirms high accuracy. (2) Residuals vs Predicted values — the random scatter around the zero line with no funnel shape or curvature confirms homoscedasticity (constant variance) and the absence of systematic bias. (3) Residual distribution histogram — the approximately Gaussian shape validates the normality assumption underlying least-squares regression, ensuring that confidence intervals and statistical tests are reliable.

## 4.6 Unseen Data Performance




The ultimate generalization test: model performance on 500 completely new cheese batches generated with a different random seed, never seen during training or validation. The left panel shows actual vs predicted scatter maintaining the tight diagonal pattern. The center shows error distribution centered near zero with no systematic bias. The right panel compares $R^2$ across train, test, and unseen sets — minimal degradation across all three confirms production readiness and that the model captures the true underlying relationship rather than memorizing training data.

# PROJECT 2

Silicon Fmax Validation — Predicting Maximum Clock Frequency

## 1 Overview

**Objective:** Apply the same linear regression fundamentals to a post-silicon validation use case — predicting the maximum stable clock frequency (Fmax) of silicon chips across voltage and temperature corners.

**Domain:** Semiconductor Engineering & Post-Silicon Validation

**Target Variable:** Fmax (MHz)

In semiconductor manufacturing, characterizing every chip at every voltage-temperature corner is prohibitively expensive. A well-trained regression model can predict Fmax for untested conditions, reducing physical characterization runs by 40–60%. This project demonstrates how the same OLS → Ridge → Lasso approach transfers directly from food science to silicon engineering.

### Engineering Value

- Predict Fmax at untested voltage/temperature corners, reducing test runs by 40–60%.
- Flag silicon lots likely to fail timing at high temperature via residual outlier detection.
- Provide fast first-pass estimates during chip bring-up before full characterization.
- Lasso's feature selection identifies redundant measurements for test plan optimization.

## 2 Datasets

### Silicon Fmax Validation Data

| Property | Value |
|---|---|
| **Records** | 2,000 samples |
| **Features** | 7 (6 numeric + 1 categorical) |
| **Target** | fmax_mhz (maximum stable clock frequency) |
| **Split** | Train 1,400 · Validation 300 · Test 300 |
| **Missing Values** | None |

## Feature Descriptions

| Feature | Description |
|---|---|
| **vdd_core** | Core supply voltage (V) |
| **junction_temp** | Junction temperature (°C) |
| **leakage_current** | Leakage current measurement (μA) |
| **ring_oscillator_speed** | Ring oscillator frequency (MHz) |
| **thermal_resistance** | Thermal resistance of the package (°C/W) |
| **ir_drop_estimate** | IR drop across the power delivery network (mV) |
| **silicon_lot_id** | Manufacturing lot identifier (categorical) |

# 3 Flowchart

Silicon Fmax Validation Pipeline
Load Fmax Dataset
Read silicon_fmax_validation_data.csv; target = fmax_mhz.
Feature Analysis (EDA)
Scatter Fmax vs VDD(+), Temp(+), leakage, ring osc, IR drop, thermal R.
Preprocess & Split
Standardize numerics, one-hot silicon_lot_id; 70/15/15 train/val/test.
Train Linear Models
Fit OLS, RidgeCV, LassoCV on processed features.
3D Cost Surface
Visualize MSE bowl over (w_vdd_w_temp) with others fixed.
L1 vs L2 Geometry
Elliptical contours; L2 sphere vs L1 diamond (sparsity).
Cross-Validation
5-fold alpha search (1e−3...1e3) selects regularization strength.
Test Evaluation
RMSE/MAE/R² on held-out test; Ridge parity check.
Unseen Silicon Lots
500 new fab samples; transform, predict, report metrics.

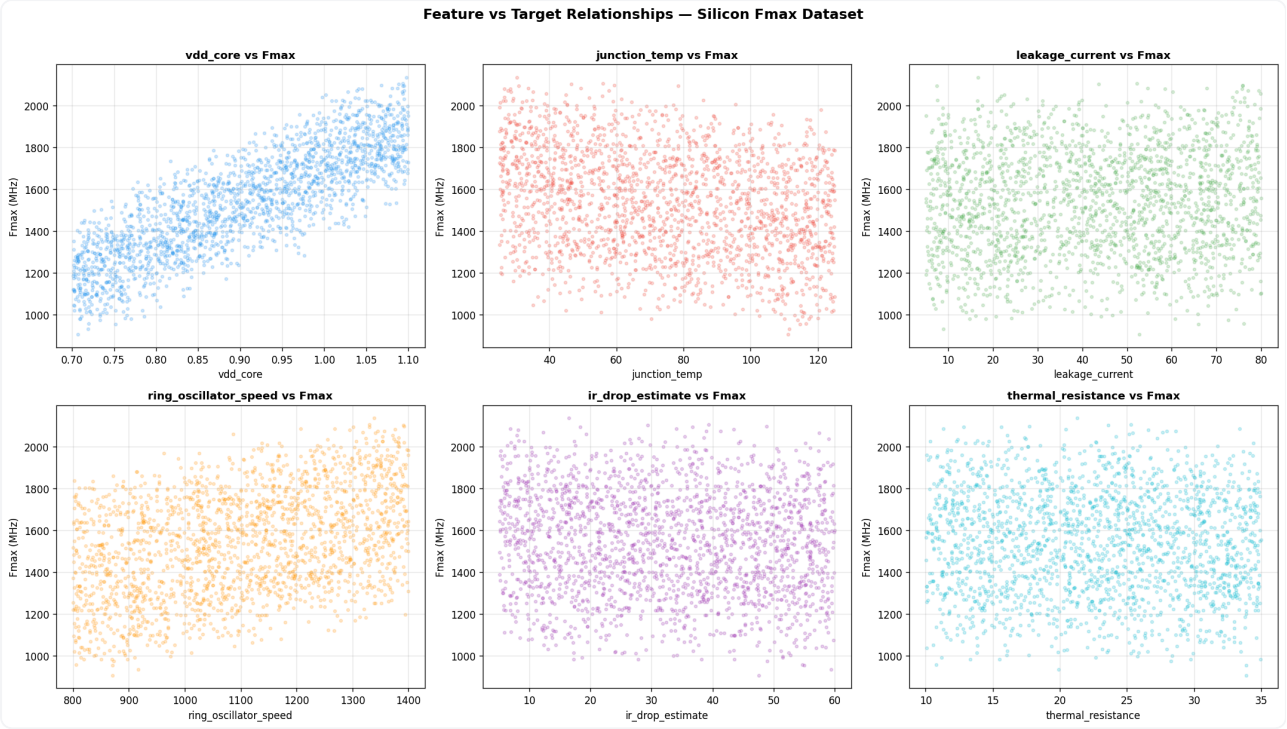
# 4 Results

All models evaluated on a held-out test set of 300 silicon samples. OLS and Ridge achieved near-identical performance at $R^2 = 0.9960$, explaining 99.6% of Fmax variance.

| Model | RMSE (MHz) | MAE (MHz) | $R^2$ |
|---|---|---|---|
| **OLS** | 15.14 | 12.09 | 0.9960 |
| **Ridge (α = 0.13)** | 15.14 | 12.09 | 0.9960 |
| **Lasso (α = 0.003)** | 15.14 | 12.09 | 0.9960 |
| **Elastic Net** | 20.56 | 16.90 | 0.9927 |

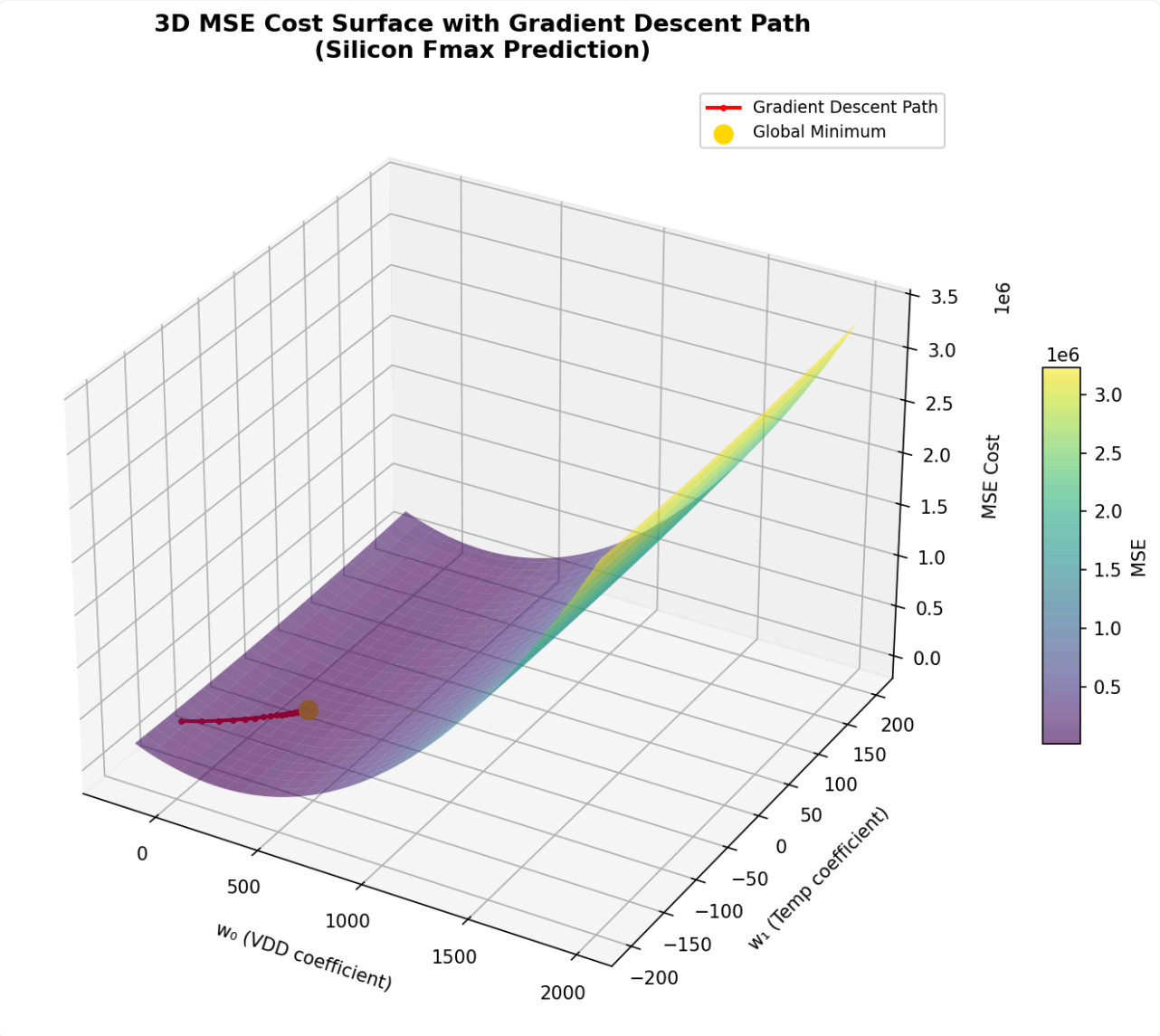
**Best Model:** Ridge — Test $R^2 = 0.9960$ · RMSE = 15.14 MHz — predicts Fmax within ±15 MHz across all voltage and temperature corners.

## 4.1 Feature Analysis



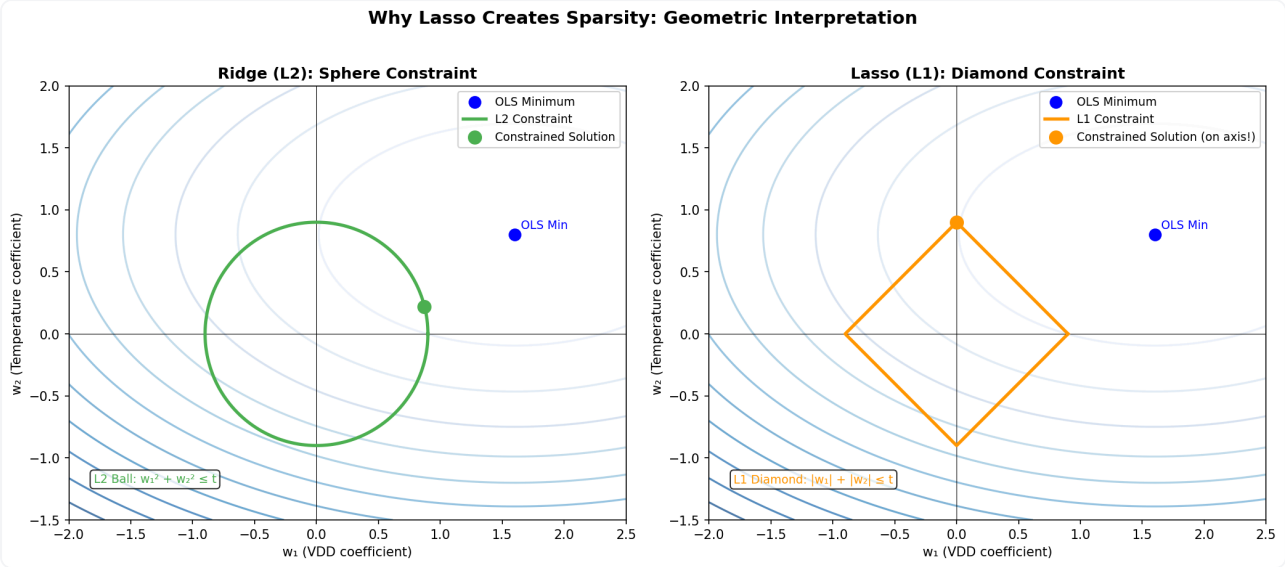

Six scatter plots mapping each silicon measurement against Fmax. A strong positive correlation with VDD core voltage is immediately visible — higher supply voltage enables higher clock speeds, as expected from circuit theory. Ring oscillator speed also shows a strong positive relationship, serving as a reliable proxy for process speed. Junction temperature exhibits a clear negative trend (higher temperature degrades timing margins), while leakage current reveals a subtle non-linear relationship tied to transistor threshold voltage variation.

## 4.2 3D Cost Surface



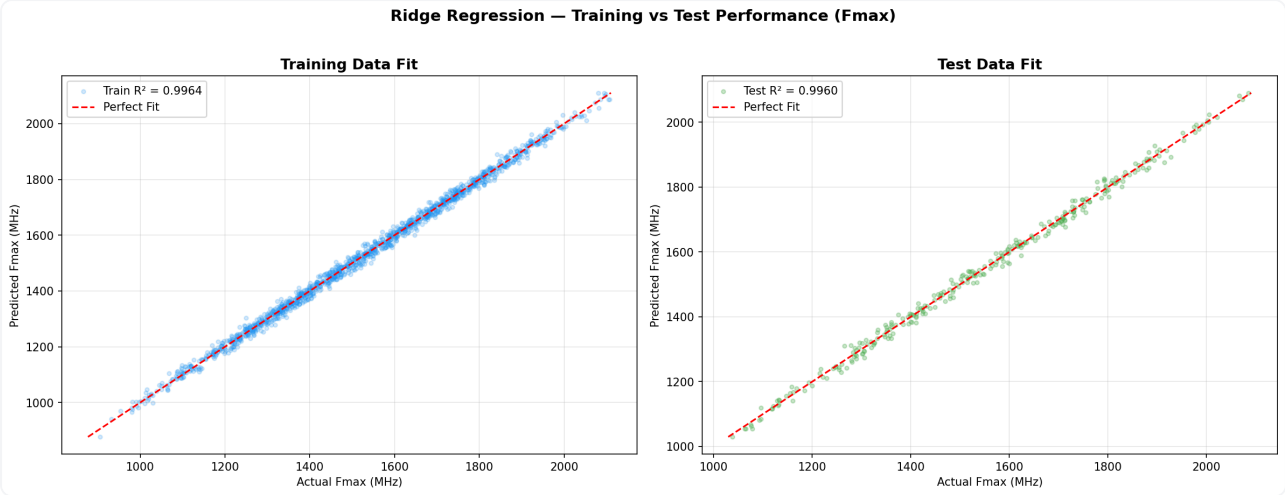

A 3D visualization of the MSE loss landscape with respect to the two most influential coefficients — VDD core voltage and junction temperature. The smooth bowl-shaped surface has a single global minimum where the optimal weights lie. The red gradient descent trajectory spirals downward from the initial random point to the minimum, demonstrating exactly how iterative optimization navigates the parameter space. The convexity of this surface guarantees that gradient descent will always find the same global optimum regardless of initialization.

## 4.3 L1 vs L2 Regularization Geometry



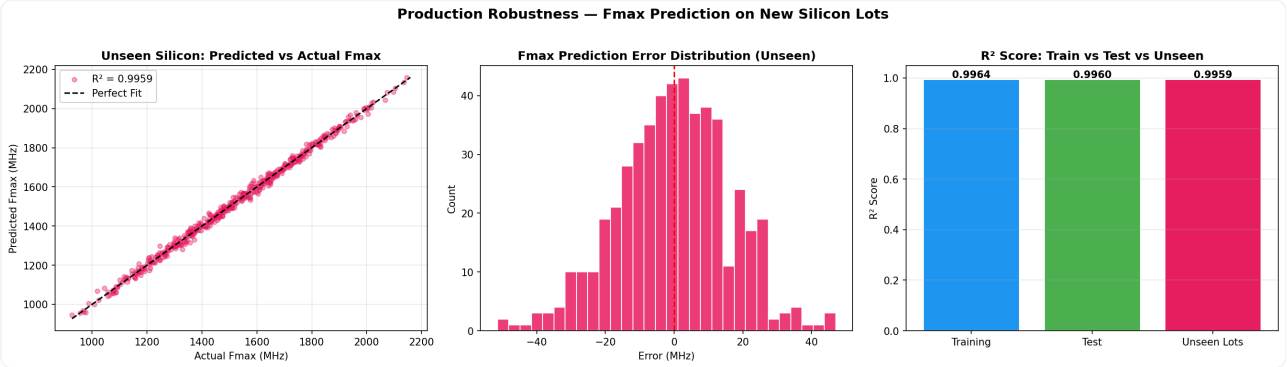

The fundamental geometric intuition behind regularization: L1 (Lasso) constrains coefficients within a diamond-shaped region, while L2 (Ridge) constrains within a circle. The elliptical contour lines represent the unpenalized MSE loss surface. The regularized solution sits where the loss contours first touch the constraint boundary — for L1, this intersection tends to occur at the diamond's corners (on the axes), where one or more coefficients equal zero. This is why Lasso naturally produces sparse models with automatic feature selection, whereas Ridge only shrinks coefficients without eliminating them.

## 4.4 Train vs Test Fit




Actual vs predicted Fmax on both training and test silicon lots, with the ideal y = x diagonal as reference. The exceptionally tight clustering with $R^2 = 0.9960$ demonstrates that the model accurately captures the underlying physics linking silicon measurements to maximum clock frequency. The consistency between training and test panels confirms no overfitting — the model learned generalizable patterns rather than memorizing specific lots.

## 4.5 Unseen Silicon Lots




Production robustness validation on 500 new silicon lots from an independent generation (different random seed, never seen during training). The left panel shows actual vs predicted Fmax maintaining the tight diagonal. The center shows prediction error distribution centered near zero with a narrow spread of ±15 MHz. The right panel compares $R^2$ across train, test, and unseen sets — virtually identical scores across all three confirm the model is ready for deployment in a real post-silicon validation workflow, reliably predicting Fmax for untested corners.

# A Tech Stack & Key Learnings

## Technology Stack

| Tool | Version | Purpose |
|---|---|---|
| **Python** | 3.11+ | Core language |
| **NumPy** | 1.24+ | Linear algebra, Normal Equation |
| **Pandas** | 2.0+ | Data manipulation |
| **scikit-learn** | 1.3+ | Ridge, Lasso, ElasticNet, CV |
| **Matplotlib** | 3.7+ | All visualizations including 3D |
| **Seaborn** | 0.12+ | Statistical plots |
| **FastAPI** | 0.100+ | REST API serving endpoint |
| **Plotly.js** | 2.x | Interactive production dashboard |

## Key Learnings

- Implementing OLS from scratch (Normal Equation + Gradient Descent) solidified the mathematical foundations that scikit-learn abstracts away — understanding *why* the closed-form solution works and when gradient descent is necessary.
- The geometric interpretation of L1 vs L2 regularization (diamond vs sphere constraint) provides the clearest explanation for why Lasso produces sparsity while Ridge does not.
- Applying the same pipeline to two completely different domains (food science and semiconductor engineering) proved that linear regression fundamentals transfer directly across industries.
- Train vs test fit analysis and unseen data simulation are essential validation steps — $R^2$ alone on the test set is insufficient to guarantee production-level generalization.
- The production Fmax dashboard with 8,000 synthetic dies across 50 lots demonstrates how a trained model integrates into a real post-silicon validation workflow.

## Repository

github.com/AIML-Engineering-Lab/001_linear_regression_engine · MIT License